

# Number of Ways of Partitioning a Rectangle

Asked 10 years, 5 months ago   Modified 10 years, 5 months ago   Viewed 1k times

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Given a rectangle of integer sidelengths  $m \times n$ , consider partitioning it into smaller rectangles also of integer sidelengths. How many such partitions are possible?

I wonder if this is just an exercise-level problem or something that hasn't been solved... it is such a simple question that was probably studied before.

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asked Apr 3, 2015 at 15:54

 finnlim

2,843   2   20   28

- ▲

Without any constraint? I mean, a vertex of a rectangle can be in the interior of a side of another rectangle? – Daniel Apr 3, 2015 at 15:59
- ▲

well i'm talking about partitioning so that should be a included possibility. If such partitioning is excluded then I believe the number of partitioning would be  $2^{(m-1)+(n-1)}$  – finnlim Apr 3, 2015 at 16:27 
- ▲

See [OEIS A116694](#) and [A333476](#) – Henry Aug 24, 2021 at 12:13

## 2 Answers

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There is a nice discussion of a more simple problem here: [Number of ways to partition a rectangle into n sub-rectangles](#) Given its complexity, I have doubts that your problem has a nice closed-formed answer.

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answered Apr 13, 2015 at 11:46

 macleginn

359   2   10



We consider the rectangle  $R_{mn} := [0, m] \times [0, n]$  as a band of "small" width  $m$ , fixed once and for all, and variable height  $n$ . Denote the number of partitions of  $R_{mn}$  into subrectangles by  $P(n)$ .

1



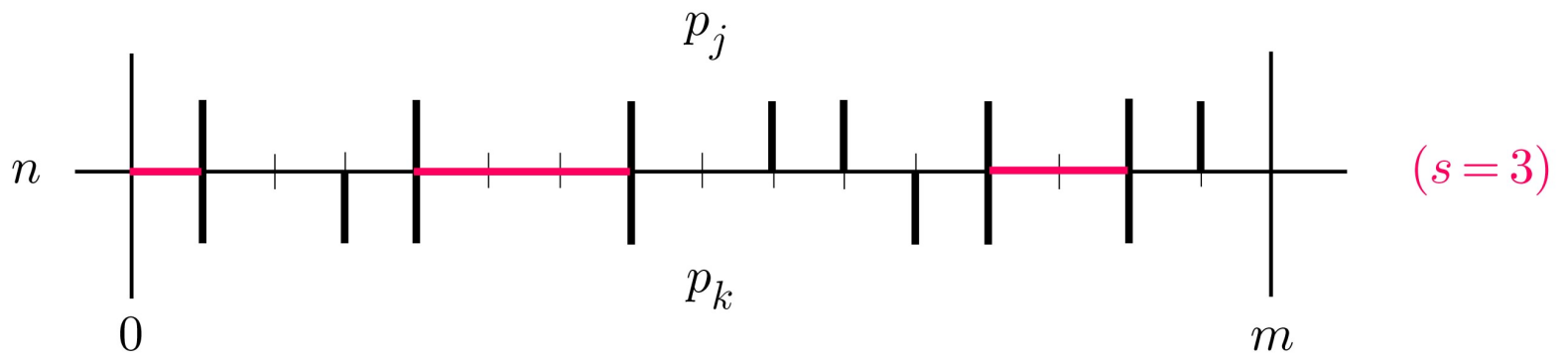
Any partition of  $R_{m,n+1}$  reduces to a partition of  $R_{mn}$  when the topmost row of little squares is chopped off. But each partition of  $R_{mn}$  can be extended to a partition of  $R_{m,n+1}$  in various ways, depending on the inner vertical edges present (i.e., used for separating two subrectangles) in the top row of  $R_{mn}$ . There are  $m - 1$  inner edges, allowing for  $N := 2^{m-1}$  different patterns of present edges. Let  $p_k$  ( $1 \leq k \leq N$ ) be the pattern defined by the binary expansion of  $k - 1$ .



Assume that pattern  $p_k$  is present in the top row of  $R_{mn}$ . Then a priori any pattern  $p_j$  can be selected for the  $(n + 1)^{\text{th}}$  row.



When the two patterns are aligned most top rectangles of  $R_{mn}$  cannot be extended upwards in an admissible way, but a certain number  $s = s_{jk} \geq 0$  of them can. For each of these we can choose to extend or not, making a total of  $2^s$  choices.



The numbers  $s_{jk}$  depend on the patterns  $p_j, p_k$  under consideration and have to be determined in a preparatory step. The  $s_{jk}$  give rise to transition matrix  $Z$  of type  $N \times N$  as follows:

$$z_{jk} = 2^{s_{jk}} \quad (1 \leq j \leq N, 1 \leq k \leq N) .$$

Denote the number of partitions of  $R_{mn}$  having pattern  $p_k$  in its top row by  $x_k(n)$ . Then

$$x_k(1) = 1 \quad (1 \leq k \leq N) .$$

We now have the recursion

$$x_j(n+1) = \sum_{k=1}^N z_{jk} x_k(n) \quad (n \geq 1) ,$$

in short:

$$\mathbf{x}(n+1) = Z \mathbf{x}(n) .$$

This finally gives

$$P(n) = \sum_{k=1}^N x_k(n) .$$

Two examples: Letting  $m = 4$  the matrix  $Z$  is given by

$$\begin{pmatrix} 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 4 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 2 & 1 & 1 & 2 & 1 \\ 1 & 2 & 2 & 8 & 1 & 2 & 1 & 4 \\ 1 & 1 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 2 & 1 & 2 & 2 & 8 & 2 & 4 \\ 1 & 1 & 2 & 1 & 2 & 2 & 8 & 4 \\ 1 & 2 & 1 & 4 & 2 & 4 & 4 & 16 \end{pmatrix}$$

and produces

$$P(10) = Z^9(1, 1, 1, \dots, 1)^\top = 10\,225\,294\,476\,962 .$$

For an  $8 \times 8$  chessboard one obtains

partitions into rectangles.

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I pretty much reached the same stage of calculation as this, but since I could not determine a very clean-cut way of producing the matrix  $Z$  so I didn't see the problem as solved. One could *program* it but a closed formula would be more convenient. But anyway, thanks for the reply! – [finnlim](#) Apr 15, 2015 at 19:39



answered Apr 10, 2015 at 17:59

[Christian Blatter](#)

233k

14

204

490

↑

Also, I haven't exactly considered the matrix formulation. This is a nice feature. – [finnlim](#) Apr 15, 2015 at 19:40

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